

# A $\mathbb{Z}_{37}$ orbit-matrix reduction and cyclotomic obstructions for the conference graph of $H(668)$

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## Abstract

A Hadamard matrix of order  $668 = 4 \cdot 167$  is the smallest order whose existence is unknown; equivalently no symmetric conference matrix  $C(334)$ , regular two-graph on 334 points, or strongly regular graph  $\text{srg}(333, 166, 82, 83)$  is known. We prove every order-37 automorphism of such a graph is necessarily fixed-point-free (a quotient-trace pincer, sharp exactly at the fixed-point-free case), and order 37 is permitted by the eigenvalue-field obstruction (unlike the group-developed Cayley case); so all order-37 symmetry reduces to a  $9 \times 9$  array of circulants over  $\mathbb{Z}_{37}$ , the strong-regularity identity becoming a per-character condition  $M_t^2 + M_t - 83I = 0$  over  $\mathbb{Z}[\zeta_{37}]$ . We then (i) completely enumerate, via a sound symmetry reduction, the 2901 admissible  $9 \times 9$  orbit matrices, the necessary skeletons of any such graph; (ii) prove by two lift-free arguments that *no* cyclotomically-structured (Paley-type) graph exists for the 6 proper nontrivial multiplier subgroups of order  $\geq 4$ ; (iii) reduce the two finest multipliers  $|H| \in \{2, 3\}$  to a per-skeleton lift we leave open, and prove a *collapse theorem* explaining the difficulty (the negation involution forces every trace-linear invariant, spectral or character-theoretic or association-scheme/Krein, to be determined by the orbit matrix, so any obstruction must be non-linear in the  $\{0, 1\}$  connection-set indicators), while confining a hypothetical order-2 or order-3 solution to  $\theta$ -multiplicity  $a = 4$  with pinned diagonal coset counts, the order-3 case to exactly four orbit-matrix classes; and (iv) document that a base-point gauge-fixing routinely used to accelerate SAT search becomes *unsound* once a multiplier constraint is imposed. Parallel-tempering evidence points to nonexistence. We release validated open-source code, the enumeration, and a Lean 4 formalization of the collapse theorem; an appendix records a negative result on square-divisor spectral filters for the equivalent Legendre pair  $\text{LP}(333)$ .

## 1 Introduction

A *Hadamard matrix* of order  $n$  is an  $n \times n$  matrix  $H$  with entries in  $\{\pm 1\}$  and  $HH^\top = nI$ . The Hadamard conjecture asserts that such a matrix exists for every  $n \equiv 0 \pmod{4}$ . The smallest order for which existence is open is  $n = 668 = 4 \cdot 167$  [1, 2]. Several classical objects are each equivalent to  $H(668)$ ; we work with two.

**Legendre pairs.** For  $x \in \{\pm 1\}^\ell$  let  $\text{PAF}_x(k) = \sum_{i \in \mathbb{Z}_\ell} x_i x_{i+k}$  be the periodic autocorrelation. A *Legendre pair* (LP) of length  $\ell$  is a pair  $u, v \in \{\pm 1\}^\ell$  with  $\text{PAF}_u(k) + \text{PAF}_v(k) = -2$  for all  $k \not\equiv 0$ . An LP of length  $\ell$  yields a Hadamard matrix of order  $2\ell + 2$  via the two-circulant-core construction [3]. For  $H(668)$  one needs  $\ell = 333 = 9 \cdot 37$ ;  $\text{LP}(333)$  is open [4, 5].

**Conference graphs.** A symmetric conference matrix  $C$  of order  $m = 334$  ( $C = C^\top$ , zero diagonal,  $\pm 1$  off-diagonal,  $C^2 = (m - 1)I$ ) yields  $H(668)$  via  $H = \begin{bmatrix} C+I & C-I \\ C-I & -C-I \end{bmatrix}$ , and  $C(334)$  is equivalent to a regular two-graph on 334 points, whose descendant is the strongly regular graph  $\Gamma = \text{srg}(333, 166, 82, 83)$  [7, 2]. The parameter set  $(333, 166, 82, 83)$  is listed as open in Brouwer's tables, and no catalogued conference-matrix construction (Paley; Mathon's  $q^2(q+2)+1$  family [8]) reaches order 334.

**Prior obstructions.** It is known that no *group-developed* (Cayley) conference graph on 333 vertices exists: its nontrivial eigenvalues are  $(-1 \pm \sqrt{333})/2$  with  $\sqrt{333} = 3\sqrt{37} \in \mathbb{Q}(\sqrt{37})$  of conductor 37, while every group of order  $333 = 3^2 \cdot 37$  has commutator subgroup inside the Sylow-37 subgroup and hence admits a character of order 3; a Cayley eigenvalue  $\chi(S)$  of order-3 character lies in  $\mathbb{Z}[\zeta_3] \not\subseteq \mathbb{Q}(\sqrt{37})$ , a contradiction (Lemma 1). Likewise the relevant prime 2 is not self-conjugate modulo 167 ( $\text{ord}_{167}(2) = 83$ ), so the Turyn–Schmidt field-descent bounds do not forbid  $H(668)$ ; the barrier is constructive, not arithmetic.

**Contributions and related work.** We (i) give a  $\mathbb{Z}_{37}$  orbit-matrix reduction of  $\Gamma$  permitted by the eigenvalue-field obstruction, distinct from the group-developed case (Section 2); (ii) completely enumerate the 2901 admissible orbit matrices and prove, by two lift-free arguments, the nonexistence of cyclotomically-structured graphs for the 6 multiplier subgroups of order  $\geq 4$  (Section 3); (iii) for the two finest multipliers  $|H| \in \{2, 3\}$ , prove a trace-linearity collapse theorem and confine a hypothetical order-2 solution, while documenting a soundness pitfall in gauge-fixed SAT search (Section 3); and (iv) record, in Appendix A, that square-divisor spectral filters do not strengthen the Kotsireas–Koutschan filter for the equivalent  $\text{LP}(333)$ . The orbit-matrix method for strongly regular graphs with a prescribed semiregular automorphism is due to Behbahani–Lam [10] and was developed for two-graphs and self-orthogonal codes by Crnković–Maksimović–Rodrigues–Rukavina [11] and, for regular two-graphs on up to 50 points, by Maksimović [12]. Our contribution is its application to this open conference parameter set: the resulting per-character condition  $M_t^2 + M_t - 83I = 0$ , the complete skeleton enumeration, the cyclotomic nonexistence theorems, and the trace-linearity collapse that delimits the method.

## 2 A $\mathbb{Z}_{37}$ orbit-matrix reduction of $\text{srg}(333, 166, 82, 83)$

**Lemma 1.** *No group of order 333 admits a  $(333, 166, 82, 83)$  partial difference set; equivalently there is no group-developed conference graph on 333 vertices.*

*Sketch.* As in the introduction: a 1-dimensional character of order 3 would force a nontrivial eigenvalue  $(-1 \pm 3\sqrt{37})/2 \in \mathbb{Z}[\zeta_3]$ , impossible since  $\mathbb{Q}(\sqrt{37}) \not\subseteq \mathbb{Q}(\zeta_3)$ .  $\square$

Lemma 1 rules out the *regular* automorphism group; it does not rule out individual automorphisms. We consider an automorphism  $\sigma$  of order 37. A priori  $\sigma$  could fix  $37c$  vertices ( $0 \leq c \leq 9$ , by orbit counting); the next proposition shows the fixed-point-free case is the only one, so the reduction below covers *all* order-37 symmetry.

**Proposition 1** (Order-37 automorphisms are fixed-point-free). *Every automorphism of order 37 of an  $\text{srg}(333, 166, 82, 83)$  acts without fixed points.*

*Proof.* Suppose  $\sigma$  fixes  $f = 37c$  vertices,  $1 \leq c \leq 8$ , and let  $s = 9 - c$  be the number of 37-orbits. Since an invariant subset of a 37-cycle is empty or everything, each fixed vertex is adjacent to all or none of each orbit; writing  $a_u$  for the number of orbits dominated by the fixed vertex  $u$ ,

its fixed-degree is  $166 - 37a_u$ , and we set  $G = \sum_u a_u$ . The partition into fixed singletons and orbits is equitable; its quotient  $Q$  has spectrum inside  $\{166, \theta, \tau\}$  with 166 simple (Perron) and  $\theta, \tau$  paired by rationality (irreducibility of  $x^2 + x - 83$ , discriminant 333 non-square), so  $\text{char}(Q) = (x - 166)(x^2 + x - 83)^b$  with  $b = (f + s - 1)/2 = 166 - 18s$ . Let  $B$  be the orbit-orbit block of  $Q$  (entries  $B_{OO'} = \text{neighbours in } O' \text{ of a vertex of } O$ ; symmetric since orbits have equal size),  $D = \text{tr } B$  and  $T$  its off-diagonal sum. Computing  $\text{tr } Q$  and  $\text{tr } Q^2$  once spectrally and once blockwise gives

$$D = 18s, \quad \sum_{O, O'} B_{OO'}^2 = 3136s - 37G = 37T - 2340s,$$

the last equality via the row sums  $\sum_{O'} B_{OO'} = 166 - \gamma_O$  ( $\gamma_O = \text{fixed vertices dominating } O$ ,  $\sum_O \gamma_O = G$ , whence  $T = 148s - G$ ). For  $s = 1$  this forces  $\sum B^2 = -2340 < 0$ , absurd. For  $2 \leq s \leq 8$ , Cauchy–Schwarz on the diagonal and off-diagonal entries separately gives  $\sum B^2 \geq 324s + T^2/(s(s-1))$ , hence

$$q(T) := T^2 - 37s(s-1)T + 2664s^2(s-1) \leq 0,$$

whose discriminant  $s^2(s-1)[1369(s-1) - 10656]$  is negative for all  $s \leq 8$  (as  $1369 \cdot 7 = 9583 < 10656$ ): no real  $T$  satisfies it. Hence  $c = 0$ . The bound is sharp: at  $s = 9$  the same system is satisfiable with slack ( $\Phi(9) = +296$ ), as witnessed by the enumerated orbit matrices. Every identity and the exhaustive integer scan over  $G$  are verified in `fpf_check.py`, including end-to-end validation on conference graphs with genuine fixed-point automorphisms (no false kills).  $\square$

By Proposition 1,  $\sigma$  partitions the vertices into 9 orbits of size 37; we identify each orbit with  $\mathbb{Z}_{37}$  so that  $\sigma$  acts as  $+1$ . The adjacency matrix becomes a  $9 \times 9$  array of circulants: writing  $D[i][j] \subseteq \mathbb{Z}_{37}$  for the connection set of block  $(i, j)$ ,  $A_{(i,a),(j,b)} = 1 \iff (b-a) \bmod 37 \in D[i][j]$ , with  $D[j][i] = -D[i][j]$  (symmetry) and  $0 \notin D[i][i] = -D[i][i]$  (no loops). Crucially the eigenvalue-field obstruction of Lemma 1 does *not* apply:  $\sqrt{37} \in \mathbb{Q}(\zeta_{37})$  (the quadratic Gauss sum), so order 37 is permitted. The reduction of a strongly regular graph to circulant blocks under a semiregular automorphism is the standard orbit-matrix method [10, 11]; our contribution is its application to this open parameter set, the resulting per-character condition, and the (non)existence results below.

**Proposition 2** (Character reduction). *Define, for  $t \in \mathbb{Z}_{37}$ , the  $9 \times 9$  matrix  $M_t[i][j] = \sum_{d \in D[i][j]} \zeta_{37}^{td}$ . Then  $A$  is  $\text{srg}(333, 166, 82, 83)$  (equivalently  $A^2 + A - 83I = 83J$ ) if and only if*

$$(t = 0) \quad R := M_0 \text{ satisfies } R^2 + R - 83I_9 = 3071J_9, \quad (t \neq 0) \quad M_t^2 + M_t - 83I_9 = 0,$$

where each  $M_t$  ( $t \neq 0$ ) is Hermitian over  $\mathbb{Z}[\zeta_{37}]$  with spectrum in  $\{(-1 \pm \sqrt{333})/2\}$ .

The integer matrix  $R = (|D[i][j]|)$ , the *orbit matrix*, is symmetric with row sums 166, even diagonal, and (from  $R^2 + R - 83I = 3071J$ ) eigenvalues 166 (once) and  $(-1 \pm \sqrt{333})/2$  (four times each); in particular  $\text{tr } R = 162$  and a Cauchy–Schwarz argument confines the diagonal entries to  $\{10, 12, \dots, 26\}$ . We verified Proposition 2 by reconstructing the Paley conference graphs  $P(9), P(25), P(49)$  in block-circulant form, confirming agreement of the direct and per-character tests.

### 3 Computational attack and results

**Orbit-matrix census.** A backtracking search with knapsack-tight pruning, made complete by a sound symmetry reduction (within each maximal equal-diagonal group of orbits, sort by connection

to a fixed orbit), enumerates *all* admissible orbit matrices up to orbit relabeling: exactly 2901 representatives, with every diagonal class exhausted. (An earlier run without the reduction had confirmed  $\geq 1678$  before several highly symmetric diagonals exceeded a  $4 \times 10^9$ -node cap.) The symmetry reduction is partial, so representatives may repeat isomorphism types; an exact canonical-form pass (lexicographic minimum over all  $9!$  relabelings) collapses the 2901 representatives to exactly 625 isomorphism classes. These form the complete set of necessary skeletons, so the lift-free arguments of Theorem 1 apply to all of them; the remaining obstruction for the fine multipliers is the *lift* to connection sets satisfying Proposition 2 (Remark 1). A representative skeleton (row sums 166,  $\text{tr} = 162$ , spectrum  $\{166, r^4, s^4\}$ ) is

$$R = \begin{pmatrix} 10 & 19 & 19 & 19 & 19 & 20 & 20 & 20 & 20 \\ 19 & 14 & 22 & 22 & 20 & 15 & 15 & 18 & 21 \\ 19 & 22 & 14 & 22 & 20 & 15 & 18 & 21 & 15 \\ 19 & 22 & 22 & 14 & 20 & 15 & 21 & 15 & 18 \\ 19 & 20 & 20 & 20 & 18 & 24 & 15 & 15 & 15 \\ 20 & 15 & 15 & 15 & 24 & 20 & 19 & 19 & 19 \\ 20 & 15 & 18 & 21 & 15 & 19 & 24 & 17 & 17 \\ 20 & 18 & 21 & 15 & 15 & 19 & 17 & 24 & 17 \\ 20 & 21 & 15 & 18 & 15 & 19 & 17 & 17 & 24 \end{pmatrix}, \quad R^2 + R - 83I_9 = 3071 J_9.$$

**Exact lift (SAT).** The lift is equivalent to the convolution system, for all  $i, j \in \mathbb{Z}_9$ ,  $g \in \mathbb{Z}_{37}$ ,

$$\sum_{i'} (D[i][i'] * D[i'][j])(g) = \mu + (k - \mu)[i=j, g=0] + (\lambda - \mu)[g \in D[i][j]],$$

which we encode in a Boolean SAT/CP model [9] (membership variables, cardinality, symmetry). For the *general* (unstructured) lift a base-point gauge-fixing removes the  $37^8$  per-orbit symmetry, and the encoding so validated reconstructs  $P(9), P(25)$ . We caution that this gauge-fixing is *unsound* once a multiplier (cyclotomic) constraint is added (Remark 1); the cyclotomic results of Theorem 1 therefore rely only on the gauge-free arguments (a),(b).

Call a  $\mathbb{Z}_{37}$ -symmetric  $\text{srg}(333, 166, 82, 83)$  *H-cyclotomic* if every connection set is a union of cosets of a multiplier subgroup  $H \leq \mathbb{Z}_{37}^*$  (the Paley-type structure). The next theorem rules these out completely for the 6 proper nontrivial subgroups of order  $\geq 4$ , by two lift-free arguments (no base-point gauge); the two finest cases  $|H| \in \{2, 3\}$  reduce to a cyclotomic lift that we leave open (Remark 1).

**Theorem 1** (No cyclotomically-structured lift, coarse multipliers). *No H-cyclotomic  $\text{srg}(333, 166, 82, 83)$  exists for  $|H| \in \{4, 6, 9, 12, 18, 36\}$ .*

- (a) *(Via the orbit matrix.) An H-cyclotomic graph forces its  $9 \times 9$  orbit matrix to have H-coset-compatible cardinalities: off-diagonal  $R[i][j] \equiv 0$  or  $1 \pmod{|H|}$  (the residue records whether  $0 \in D[i][j]$ ), and diagonal  $R[i][i] \equiv 0 \pmod{|H|}$  if  $-1 \in H$  else  $\pmod{2|H|}$ . No integer matrix meets these together with the orbit-matrix identity  $R^2 + R - 83I = 3071J$ : an independent exhaustive backtracking enumeration over the coset-compatible alphabet finds none for each of the six orders (a complete-search certificate trusting no solver; each search is exhausted in well under a second). Hence no such graph exists, for any orbit matrix.*
- (b) *(Independent confirmation,  $|H| \in \{12, 18, 36\}$ .) The stronger orbit-matrix-free system (every block H-invariant, the graph 166-regular, R unconstrained) is directly UNSAT.*

Both arguments are lift-free, hence independent of any base-point gauge. They were validated by reconstructing the Paley graphs  $P(9), P(25), P(49)$  (orbit-matrix-free) and  $P(13), P(17)$  (orbit matrix).

**Remark 1** (The fine multipliers  $|H| \in \{2, 3\}$  are open). The orbit-matrix argument (a) cannot reach  $|H| \in \{2, 3\}$ : order 2 imposes no congruence (every  $R$  is compatible), and order-3-compatible orbit matrices *do* exist. The compatibility congruences (diagonal  $\equiv 0 \pmod 6$ , off-diagonal  $\pmod 3 \in \{0, 1\}$ ) select exactly 28 of the 2901 representatives, and an exact canonical-form pass shows these constitute exactly *four* isomorphism classes, all with diagonal  $(12, 12, 18, 18, 18, 18, 18, 24, 24)$  and exactly 18 unordered block pairs containing 0; the four classes consist of one complementary pair and two self-complementary classes. (An independent mod-3-restricted enumeration without the equal-diagonal reduction finds 960 matrices; they collapse to the same four classes, a consistency check.) Both fine cases reduce to the per-orbit-matrix cyclotomic lift over the complete set of admissible orbit matrices: four classes for order 3, all 625 for order 2. We do not resolve this lift. A base-point gauge-fixing (forcing  $0 \in D[0][i]$ ) that greatly accelerates the *general* lift is *unsound* under a multiplier constraint: a multiplier-invariant set shifted by  $c \neq 0$  is no longer invariant (as  $(\text{mult} - 1)c = 0 \Rightarrow c = 0$ ), so the cyclotomic ansatz has no base-point freedom, and forcing  $0 \in D[0][i]$  wrongly rejects even the genuine negation-invariant Paley graph  $P(25)$ . Without that gauge, neither the element-level nor a coset-level SAT encoding decides a single  $|H| \in \{2, 3\}$  instance within 30 minutes. We therefore leave  $|H| \in \{2, 3\}$  open. The parallel-tempering evidence below points to nonexistence, and Propositions 3 and 4 confine a hypothetical solution to  $\theta$ -multiplicity  $a \in \{4, 5\}$  ( $a = 4$  after Lemma 2) with pinned diagonal coset counts; but a proof remains open.

We also quantify why *exhaustive* search cannot certify these cases. We implemented a coset-level backtracking solver (validated for soundness and completeness against an independent definitional enumeration on ten small instances, and against the Paley anchors) with exact per-class convolution propagation, adaptive autocorrelation interval bounds, Cauchy-interlacing spectral pruning at every node, the pinning of Proposition 4, and a sound decimation symmetry break. The structural obstacle is informational: block autocorrelations satisfy  $\text{conv}(D, -D)(g) = \text{conv}(D, -D)(-g)$ , so the flat-autocorrelation system of each orbit row imposes only 6 independent exact conditions on its 8 off-diagonal blocks ( $\sim 10^{2.8}$  admissible coset-unions each), leaving  $\sim 10^{15}$  admissible single-row configurations per diagonal assignment; an order-3 skeleton admits  $\sim 10^7$  diagonal assignments (after pinning and the decimation break). The contradiction mass lives in the cross-row equations, which close only when both rows are complete; at the measured  $2.8 \times 10^7$  nodes/second the certificate tree is out of reach by many orders of magnitude. Conflict-driven learning is the only complete method we know that could compress such a certificate; a calibration sweep of the order-3 case (the class-level model above with the pinning of Proposition 4 and  $a = 4$ , an encoding validated on the Paley anchors and on brute-force-confirmed empty instances) returned UNKNOWN on every one of the four skeleton classes at 2–2.5 hours of 8-worker CP-SAT per instance, so closure appears to require either proof-logged SAT with stronger in-solver algebraic propagation, or a genuinely new non-linear invariant.

**Unstructured lift (parallel tempering).** For the general (unstructured) lift we ran a batched GPU parallel-tempering search minimizing  $\sum_{t \neq 0} \|M_t^2 + M_t - 83I\|_F^2$ , which is 0 iff the lift succeeds. The search is validated: on the  $\text{srg}(25, 12, 5, 6)$  skeleton it drives the objective to 0 (a verified lift) within 18 steps. On  $\text{srg}(333, 166, 82, 83)$ , across  $\approx 450$  of the 634 skeletons with 8192 chains and replica exchange per skeleton ( $\approx 3.7 \times 10^6$  chains total), the objective plateaued at  $\approx 3.9 \times 10^5$  with no lift found.

**The fine multipliers: sharper ceilings.** Both open cases  $|H| \in \{2, 3\}$  (Remark 1) admit a sharp structural analysis, recorded here: how far each can be pushed, and why they resist. We begin with order 2. Since  $-1$  is a quadratic residue mod 37, negation-invariance of every block makes each

character matrix  $M_t$  a *real* symmetric  $9 \times 9$  matrix with eigenvalues in  $\{\theta, \tau\}$ ,  $\theta, \tau = (-1 \pm 3\sqrt{37})/2$ ; write  $a_t$  for the  $\theta$ -multiplicity of  $M_t$ . The Galois group  $\mathbb{Z}_{37}^*/\{\pm 1\}$  permutes the  $M_t$  and acts on  $\sqrt{37}$  through the Legendre symbol, forcing  $a_t = a$  on the quadratic residues and  $9 - a$  on the non-residues for a single integer  $a$ .

**Proposition 3** (Order-2 multiplicity reduction). *For a negation-invariant  $\mathbb{Z}_{37}$ -symmetric  $\text{srg}(333, 166, 82, 83)$ , the common residue-character  $\theta$ -multiplicity satisfies  $a \in \{4, 5\}$ .*

*Proof sketch.* Expanding  $\text{tr } M_1 = a\theta + (9-a)\tau$  in the integral basis  $\eta_g = \zeta_{37}^g + \zeta_{37}^{-g}$  via the quadratic Gauss sum  $\sqrt{37} = \sum_{g=1}^{18} \left(\frac{g}{37}\right) \eta_g$  identifies the diagonal pair-counts  $N_g := \#\{i : \{g, -g\} \subseteq D[i][i]\}$  as  $N_g = 3a - 9$  on the nine residue pairs and  $N_g = 18 - 3a$  on the nine non-residue pairs. Non-negativity  $0 \leq N_g \leq 9$  gives  $a \in \{3, 4, 5, 6\}$ . For  $a \in \{3, 6\}$  the counts are extremal  $\{0, 9\}$ , forcing every diagonal block to equal  $Q^\mp$  (all non-residues, resp. all residues) and hence the orbit matrix to have constant diagonal  $(18, \dots, 18)$ ; an exhaustive backtracking search, complete under the  $S_9$  orbit-relabeling symmetry ( $1.9 \times 10^8$  nodes, corroborated to  $9 \times 10^{10}$  nodes by an independent reduction), finds *no* admissible orbit matrix with constant diagonal 18. Hence  $a \in \{4, 5\}$ . The  $N_g$  identity was validated on the negation-invariant graphs  $P(13)$  and the 25-orbit Paley graph of  $\text{GF}(5^3)$ .  $\square$

The same Galois-pinning argument applies verbatim to the other open multiplier, with cosets of  $H = \{1, 10, 26\}$  (the cube roots of unity mod 37) in place of  $\pm$  pairs. Since  $H \leq Q^+$  (as  $10 = 11^2$ ), each of the 12 cosets of  $H$  is wholly residue or non-residue (6 each), the Gauss periods  $\eta_c = \sum_{x \in c} \zeta_{37}^x$  form an integral basis of the degree-12 subfield  $\mathbb{Q}(\zeta_{37})^H$ , and  $1 = -\sum_c \eta_c$ ,  $\sqrt{37} = \sum_c \chi(c) \eta_c$ .

**Proposition 4** (Order-3 multiplicity reduction). *For an order-3-cyclotomic  $\mathbb{Z}_{37}$ -symmetric  $\text{srg}(333, 166, 82, 83)$ , with  $a$  the  $\theta$ -multiplicity of  $M_1$  and  $n_c := \#\{i : c \subseteq D[i][i]\}$  the diagonal coset counts, one has*

$$n_c = 3a - 9 \quad \text{on the six residue cosets,} \quad n_c = 18 - 3a \quad \text{on the six non-residue cosets,}$$

*hence  $a \in \{3, 4, 5, 6\}$ ; and  $a \in \{3, 6\}$  is impossible (it forces every diagonal block to be the union of all non-residue, resp. all residue, cosets, so the orbit matrix has constant diagonal 18, contradicting the same exhaustive search as in Proposition 3). Hence  $a \in \{4, 5\}$ : each residue coset lies in exactly  $3a - 9 \in \{3, 6\}$  of the nine diagonal blocks and each non-residue coset in  $18 - 3a$ .*

*Proof sketch.* Each diagonal block is a disjoint union of  $H$ -cosets ( $-1 \in Q^+$  pairs cosets  $c \leftrightarrow -c$  within a quadratic class), so  $\text{tr } M_1 = \sum_c n_c \eta_c$ . Equating with  $a\theta + (9-a)\tau = -\frac{9}{2} + \frac{3(2a-9)}{2}\sqrt{37}$  expanded in the period basis gives  $n_c = \frac{9+3(2a-9)\chi(c)}{2}$ , and the claims follow as before. The period-basis identities and the count formula were validated numerically (nonsingularity of the Galois matrix of periods, the Gauss-sum expansion on every conjugate, exact recovery of planted coset counts, and the 1-orbit anchor  $P(13)$  with  $H = \{1, 3, 9\}$ , where  $D = Q^+(13)$  realizes  $n_c = (1 + \chi(c))/2$ ).  $\square$

**Lemma 2** (Decimation normalization). *For  $|H| \in \{2, 3\}$  and  $u \in \mathbb{Z}_{37}^*$ , replacing every block by  $uD[i][j]$  yields an isomorphic  $H$ -cyclotomic graph with the same orbit matrix and  $\theta$ -multiplicity  $a$  if  $\chi(u) = 1$ ,  $9 - a$  if  $\chi(u) = -1$ . Hence in Propositions 3 and 4 one may take  $a = 4$  without loss of generality. Similarly complementation preserves  $\mathbb{Z}_{37}$ -symmetry and the cyclotomic structure and maps  $a \mapsto 9 - a$  while complementing the orbit matrix ( $R[i][j] \mapsto 37 - R[i][j]$  off-diagonal,  $R[i][i] \mapsto 36 - R[i][i]$ ); for order 3 it interchanges the two complementary skeleton classes and fixes the two self-complementary ones.*

*Proof.*  $(i, g) \mapsto (i, u^{-1}g)$  is an isomorphism from the decimated graph to the original;  $u \cdot$  preserves unions of  $H$ -cosets and cardinalities; and  $M'_t = M_{tu}$ , so the Galois pinning gives  $a' = a_u = a$  or  $9 - a$  according to  $\chi(u)$ .  $\square$

That the order-2 case resists our tools is structural, not accidental: the standard linear-algebraic and character-theoretic obstructions are all *trace-linear*, and they collapse.

**Theorem 2** (Trace-linearity collapse). *Let  $A$  be the adjacency matrix of an order-2-cyclotomic  $\mathbb{Z}_{37}$ -symmetric  $\text{srg}(333, 166, 82, 83)$  with orbit matrix  $R$ , and let  $\varphi \in \{0, 1\}^{333 \times 333}$  be the permutation matrix of the negation involution  $(i, g) \mapsto (i, -g)$ . Then:*

- (i)  $\varphi^2 = I$ ,  $\varphi A = A\varphi$  (so  $\varphi$  is an automorphism of  $A$ ), and  $\varphi J = J\varphi = J$ ;
- (ii) the unital algebra  $\mathcal{A} = \langle A, \varphi, J \rangle \subseteq M_{333}(\mathbb{Q})$  is spanned by  $\{I, A, J, \varphi, \varphi A\}$ , so  $\dim \mathcal{A} \leq 5$ ;
- (iii) the trace of every element of  $\mathcal{A}$  is a  $\mathbb{Q}$ -linear combination of  $\text{tr } I = 333$ ,  $\text{tr } A = 0$ ,  $\text{tr } J = 333$ ,  $\text{tr } \varphi = 9$ , and  $\text{tr}(\varphi A) = \text{tr } R = 162$ , all fixed by the parameters and  $R$ .

Consequently every spectral, character-theoretic, involution-graded, or quotient/Krein invariant of the graph, each of which is the trace of a word in  $A, \varphi, J$ , is determined by  $R$  alone. No such invariant can distinguish a hypothetical order-2-cyclotomic solution from the orbit-matrix data, hence none can obstruct existence: any obstruction must be non-linear in the  $\{0, 1\}$  connection-set indicators.

*Proof.* (i) Each block  $D[i][j]$  is negation-invariant, so  $A_{(i,-g),(j,-h)} = [(-h) - (-g) \in D[i][j]] = [h - g \in -D[i][j]] = [h - g \in D[i][j]] = A_{(i,g),(j,h)}$ ; thus  $\varphi A \varphi^{-1} = A$ , and  $\varphi = \varphi^{-1}$  gives  $\varphi A = A\varphi$ . Each row of  $\varphi$  has a single 1, so  $\varphi J = J = J\varphi$ . (ii) From  $A^2 + A - 83I = 83J$  we get  $A^2 = 83I - A + 83J$ , and 166-regularity gives  $AJ = JA = 166J$ ; by induction  $A^n \in \text{span}\{I, A, J\}$  for all  $n$ . Since  $\varphi^2 = I$  and  $\varphi$  commutes with  $A$  and  $J$  (with  $\varphi J = J$ ), every word in  $A, \varphi, J$  equals  $A^n \varphi^\varepsilon$  for some  $n \geq 0, \varepsilon \in \{0, 1\}$ , which lies in  $\text{span}\{I, A, J\} \cdot \{I, \varphi\} = \text{span}\{I, A, J, \varphi, \varphi A\}$  (using  $J\varphi = J$ ); and  $A\varphi = \varphi A$ . (iii)  $\text{tr } A = 0$  (no loops);  $\text{tr } \varphi$  counts fixed points  $(i, g)$  with  $g \equiv -g$ , i.e.  $g = 0$ , one per orbit, so 9; and  $\text{tr}(\varphi A) = \sum_{i,g} A_{(i,-g),(i,g)} = \sum_{i,g} [2g \in D[i][i]] = \sum_i |D[i][i]| = \text{tr } R = 162$ , since  $g \mapsto 2g$  permutes  $\mathbb{Z}_{37}$ . Linearity of the trace finishes the claim.  $\square$

**Remark 2** (Formal verification). Theorem 2 and the moment collapse it generalizes are formalized in Lean 4 with Mathlib (in `h668lean/` in the repository): the proof terms carry no `sorry` and depend only on the standard axioms `propext`, `Classical.choice`, `Quot.sound`. The supporting arithmetic (83 not self-conjugate mod 37, and  $a \in \{3, 4, 5, 6\}$ ) is likewise machine-checked.

**Remark 3** (Why character and multiplier methods are inert). Two consequences sharpen Theorem 2. (a) In the group ring  $\mathbb{Z}[\mathbb{Z}_{37}]$  (generator  $X$ ) the strong-regularity identity reads  $B^2 + B - 83I_9 = 83T E_9$  with  $B = (B_{ij})$  the block-indicator matrix,  $E_9$  all-ones, and  $T = \sum_g X^g$  supported on the principal character; so under any nonprincipal character  $X \mapsto \zeta_{37}^t$  the inhomogeneous term vanishes and one is left with the homogeneous  $M_t^2 + M_t - 83I = 0$ , whose Gauss-/Jacobi-sum evaluations reduce to cardinalities already fixed by  $R$ . (b) The numerical multiplier  $83 \equiv 9 \pmod{37}$  has odd order  $\text{ord}_{37}(83) = 9$  with  $-1 \notin \langle 83 \rangle$ , so 83 is not self-conjugate mod 37 and the Turyn–Schmidt field-descent / multiplier bounds impose nothing, the same arithmetic as the Paley graph  $P(13)$ , which exists. In particular the moment hierarchy collapses: as  $A$  satisfies a quadratic,  $\text{tr}(A^k)$  is an affine function of  $\text{tr } A$  for every  $k$ , so no power-sum (eigenvalue-multiplicity) count beyond  $R$  is available. (c) A further instance: reducing the  $g = 0$  (“zero-class”) convolution equations modulo  $|H|$ , where coset–coset intersections vanish, yields conditions on the pattern  $Z = (R \bmod |H|)$  of

blocks containing 0, namely  $(Z^2)_{ij} + Z_{ij} \equiv 83 - 83 \cdot 37 [i=j] \pmod{|H|}$ ; but  $Z \equiv R$ , so this is exactly the orbit-matrix identity  $R^2 + R - 83I = 3071J$  reduced mod  $|H|$ , automatically satisfied by every admissible skeleton. The candidate “zero-pattern obstruction” is the orbit matrix in disguise.

Thus the order-2 obstruction, if any, lives entirely in the non-linear  $\{0, 1\}$ -realizability of the connection sets: the lift of Remark 1, which we leave open.

## 4 Discussion

We distinguish what is proven from what is evidence. *Proven*: Proposition 5 (square-divisor filters do not help LP(333)); Proposition 2 (the reduction); the complete enumeration of the 2901 admissible orbit matrices (625 isomorphism classes); Theorem 1 (no cyclotomically-structured  $\mathbb{Z}_{37}$  lift for the 6 multiplier subgroups of order  $\geq 4$ , by two lift-free arguments); Propositions 3 and 4 with Lemma 2 (a hypothetical order-2 or order-3 solution is confined to  $\theta$ -multiplicity  $a = 4$  with pinned diagonal coset counts, the order-3 case to exactly four skeleton classes, and both resist every trace-linear obstruction). *Open*: the cyclotomic lift for the two finest multipliers  $|H| \in \{2, 3\}$  (Remark 1), which sound SAT does not decide in feasible time. *Evidence*: the parallel-tempering plateau is a strong indication, from a search demonstrably able to find lifts when they exist, that no unstructured  $\mathbb{Z}_{37}$ -symmetric lift exists either, though it is not a proof. Taken together with Lemma 1, this makes a  $\mathbb{Z}_{37}$ -symmetric  $\text{srg}(333, 166, 82, 83)$  unlikely, and focuses any future construction of  $H(668)$  on (a) genuinely asymmetric conference graphs, (b) automorphisms of other orders (for which the orbit-matrix block size grows), or (c) the Legendre-pair / cocyclic routes [13], where Appendix A clarifies which spectral tools do and do not help.

## Reproducibility

All code (the  $\mathbb{Z}_{37}$  reduction and verifier, the complete orbit-matrix enumerator with its sound symmetry reduction, the orbit-matrix coset-compatibility and orbit-matrix-free cyclotomic checks, the exact and coset-level SAT lifts together with the validation exhibiting the multiplier gauge-fixing as unsound, the GPU batched parallel-tempering search, and the conference-to-Hadamard emitter), together with its validation (reconstruction of the Paley graphs  $P(9), P(25), P(49)$ , the conference-to-Hadamard chain on  $H(12), H(28), H(36)$ , and recovery of a genuine lift on the  $\text{srg}(25, 12, 5, 6)$  skeleton) is openly available at <https://github.com/maceoCK/h668-z37> and archived on Zenodo (DOI to be assigned on release). The verifiable certificates are included: the 2901 enumerated orbit matrices (each checkable against  $R^2 + R - 83I = 3071J$  by the bundled verifier), their 625 canonical isomorphism-class representatives and the four order-3-compatible classes (exact lexicographic-minimum canonicalization over all  $9!$  relabelings), the independent coset-compatible enumeration giving count 0 for each  $|H| \in \{4, 6, 9, 12, 18, 36\}$  (Theorem 1(a)), the numerical validation of the period-basis pinning (Propositions 3, 4), and the exhaustive coset-level solver of Remark 1 with its soundness-and-completeness cross-validation harness. The repository also includes a Lean 4 / Mathlib formalization of Theorem 2 and its supporting arithmetic, machine-checked free of `sorry`.

## A A square-divisor spectral filter for LP(333) adds no pruning

The Legendre-pair route to  $H(668)$  needs an LP(333); we record a self-contained negative result on one natural filter for it, independent of the orbit-matrix development above. Write  $\text{PSD}_x(s) =$

$|\sum_j x_j \omega^{js}|^2$ ,  $\omega = e^{2\pi i/\ell}$ . Summing the LP condition gives the flat-spectrum form

$$\text{PSD}_u(s) + \text{PSD}_v(s) = 2\ell + 2 \quad (s \not\equiv 0), \quad \text{PSD}_u(0) + \text{PSD}_v(0) = 2, \quad (1)$$

so the row sums are  $\pm 1$ ; for  $\ell = 333$  the constant is  $2\ell + 2 = 668$ .

For  $d \mid \ell$  the  $d$ -compression  $c_r^{(d)} = \sum_{j \equiv r \pmod{d}} x_j$  satisfies  $\text{PAF}_{c^{(d)}}(k) = \sum_{i \equiv k \pmod{d}} \text{PAF}_x(i)$  [6], and  $\text{PSD}_x(\ell/d)$  depends only on  $c^{(d)}$ .

**Cube root (mod 3).** At  $s = \ell/3$ , write  $\gamma = c^{(3)}(u)$ , with  $\gamma_r$  odd (a sum of  $\ell/3 = 111$  values  $\pm 1$ ). Then  $\text{PSD}_u(\ell/3) = |\gamma_0 + \gamma_1 \zeta_3 + \gamma_2 \zeta_3^2|^2 = \gamma_0^2 + \gamma_1^2 + \gamma_2^2 - \gamma_0 \gamma_1 - \gamma_1 \gamma_2 - \gamma_2 \gamma_0$ , the Eisenstein norm. Since the  $\gamma_r$  are odd and  $1 + \zeta_3 + \zeta_3^2 = 0$ , this value is divisible by 4:  $\text{PSD}_u(\ell/3) = 4(x^2 + xy + y^2)$  with  $x = (\gamma_0 - \gamma_1)/2$ ,  $y = (\gamma_1 - \gamma_2)/2 \in \mathbb{Z}$ . With (1) this gives the necessary condition

$$\frac{\ell+1}{2} = 167 = L_u + L_v, \quad L_u, L_v \text{ Loeschian (values of } x^2 + xy + y^2\text{)}. \quad (2)$$

A short enumeration recovers exactly the Kotsireas–Koutschan spectrum [4]: 16 admissible values of  $\text{PSD}_u(\ell/3)$ , all  $\equiv 4 \pmod{12}$ , forming 8 complementary Loeschian pairs summing to 167.

**Ninth root (mod 9): the square-divisor refinement.** Since  $9 \mid 333$ , one may also evaluate at a primitive 9th root  $s = \ell/9$ . With  $c = c^{(9)}(u)$  (entries odd), the same parity argument gives  $\text{PSD}_u(\ell/9) = 4|E(\zeta_9)|^2$  for  $E \in \mathbb{Z}[\zeta_9]$ , a totally non-negative element of the cubic field  $K = \mathbb{Q}(\zeta_9)^+ = \mathbb{Q}(\cos \frac{2\pi}{9})$ , and (1) reads  $167 = N_u + N_v$  in  $\mathcal{O}_K$  with  $N_u, N_v$  totally non-negative. This is a genuine cubic-field condition, and no published filter uses a prime-power (rather than prime) frequency. One might hope it strengthens (2). It does not.

**Proposition 5** (Square-divisor filter adds no pruning). *For every length  $\ell \in \{27, 45, 63\}$  (i.e.  $\ell = 9m$  with  $m \in \{3, 5, 7\}$ ), every mod 3-admissible state  $(\gamma, \gamma')$  (a pair of odd triples with row sums  $\pm 1$  and  $\sum \gamma_r^2 + \sum \gamma_r'^2 = 4m + 2$ ) is the 3-compression of a valid length-9 compressed Legendre pair  $(c, c')$ . Hence the ninth-root condition is implied by the cube-root condition and provides no additional pruning of the spectrum.*

*Verification.* For each  $\ell$  we enumerate all length-9 integer pairs  $(c, c')$  with odd entries in  $[-m, m]$ , row sums  $\pm 1$ , and  $\text{PAF}_c(k) + \text{PAF}_{c'}(k) = -2m$  ( $k \neq 0$ ),  $= 2\ell - 2m + 2$  ( $k = 0$ ); we form the set of 3-compressions  $(c^{(3)}, c'^{(3)})$  and check it equals the full set of mod 3-admissible states. The counts are:  $\ell = 27$ : 250,128 pairs, all 288 states realized;  $\ell = 45$ : 1,163,808 pairs, all 288 realized;  $\ell = 63$ : 9,385,956 pairs, all 756 realized. In every case the number of mod 3-admissible states that fail to lift is 0.  $\square$

**Remark 4.** The case  $\ell = 45$  has  $(\ell + 1)/2 = 23$ , prime and  $\equiv 2 \pmod{3}$  (the same arithmetic type as  $167 = (333 + 1)/2$ ), so the negative result is tested on a length representative of 333, not merely on small toys. Structurally, the 9-compression space is enormous and is a weak filter; the cube-root norm already captures all spectral information available at the 3-part. We conclude that effort on LP(333) should not be directed at square-divisor spectral filters.

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